The Bayesian Algorithm

Lancaster, Ch. 1 — with Appendix 1 and a bridge to Ch. 2

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An Introduction to Modern Bayesian Econometrics (2004) — reading group draft

How to use this deck

Orientation

Three sources are braided together throughout:

- **Lancaster, Ch. 1** — *The Bayesian Algorithm*. The spine. Emphasis falls on §1.3 (the algorithm) and §1.4 (its components), which is where the whole book actually lives.
- **Lancaster, Appendix 1** — *A Conversion Manual*. We start here, out of order, on purpose.
- **Giocoli (2013)**, *From Wald to Savage* — the history that explains why the argument in Appendix 1 is a century old and still unfinished.

Conventions in this deck

- Slides marked **Socratic prompt** are for discussion, not for answering quickly.
- `Julia` blocks are runnable with `Distributions.jl`, `Random`, `Statistics`.
- Instructor notes are in `\note{}` blocks — uncomment `\setbeameroption{show notes}` in the header to project them on a second screen.

Where we are going

1. **Orientation** — what an econometric analysis *is* (§1.1–1.2)
2. **Appendix 1** — the frequentist/Bayesian contrast, in detail, Socratically
3. **The algorithm** — Bayes' theorem and Algorithm 1.1 (§1.3)
4. **The likelihood** — the *information transfer function* (§1.4.1)
5. **Exchangeability** — de Finetti, and why you never needed "iid"
6. **The prior** (§1.4.2) and **the posterior** (§1.4.3)
7. **When conjugacy fails** — Monte Carlo and ABC
8. **Posterior prediction** — the payoff
9. **Decisions** (§1.4.4) — econometrics vs. decision theory, with Wald
10. **Bridge to Ch. 2** — value at risk, expected shortfall, and the predictive distribution

Part I — Orientation (§1.1–1.2)

Lancaster's own claim about this chapter

"An algorithm is a set of rules for doing a calculation. The Bayesian algorithm is a set of rules for using evidence (data) to change your beliefs. ...If this explanation is successful the reader may then put down the book and start doing Bayesian econometrics, for the rest of the book is little more than illustrative examples and technical details."

- Chapter 1 and, to a lesser extent, Chapter 2 are **the** book.
- Everything after Ch. 2 — regression, MCMC, panels, IV, time series — is the same five steps, run on harder models.
- So: if we get Ch. 1 right, the rest of the semester is bookkeeping.

Lancaster (2004), Ch. 1 opening paragraph.

§1.1 What an econometric analysis is

An econometric analysis is **the confrontation of an economic model with evidence**.

- A model asserts a relation among economic variables: $C = \alpha + \beta Y$.
- It contains **potential data** (C, Y) and **parameters** $(\theta = (\alpha, \beta) \in \Theta)$ that are not directly observable.
- Any value of θ defines a **structure**: $\theta = (10, 0.9)$ gives $C = 10 + 0.9Y$.
- The model is the *set* of structures, indexed by θ.

The two questions

1. Is the model consistent with the evidence?
(model criticism $\rightarrow$ Chapter 2)

2. Given that it is, what are the probabilities of the different structures?
(inference $\rightarrow$ Chapter 1)

Practice asks these **in reverse order**: presume consistency, then find the probable structures.

Lancaster §1.1.

§1.2 Econometrics is not statistics

Statistical analysis is largely descriptive: reduce the complexity of a set of numbers to something the mind can hold. (Fisher, 1922: "the object of statistical methods is the reduction of data.")

Econometrics is primarily about the *behavior of economic agents and their interactions in markets*. Data analysis is secondary to that concern.

That primacy dictates the class of models worth considering:

- markets can be in, or near, **equilibrium**;
- agents are presumed to be **optimizing** some objective function;
- agents often **know things the econometrician does not**.

These are what make an econometric result *interpretable to an economist* and give parameters solid meaning.

Lancaster §1.2. Hold onto this slide — we return to it in Part X on decision theory.

Socratic warm-up

Socratic prompt

Lancaster says the econometrician's parameters have "solid meaning" because they come out of a theory about optimizing agents.

1. If θ is defined only inside a theory — "the elasticity of substitution," "the coefficient of risk aversion" — in what sense can it be *true*?
2. A statistician and an econometrician fit the identical regression to the identical data. What, if anything, is different about their conclusions?
3. Does the distinction survive contact with machine learning, where the model is chosen for predictive performance and has no agents in it at all?

Part II — Appendix 1: A Conversion Manual

Why we start at the back of the book

Appendix 1 is a six-page comparative guide to the two modes of inference. We read it **first** because:

- Nearly everyone in the room was trained frequentist. The default vocabulary ("distribution" means *sampling* distribution; "standard error" means *SD of the sampling distribution*) is already installed.
- Lancaster's own framing: "So overwhelming is this approach that it is often *identified* with econometrics." "
- Appendix 1 is not an argument for Bayes. It is a **contrast**: what changes, precisely, and what does not.

Key idea
Appendix 1 is the map of the conversion. Chapter 1 is the machine you convert to.

Lancaster, Appendix 1, opening.

A1.1 The frequentist approach, stated fairly

1. Envisage a **population** of economic agents; the data are a **sample** from it.
2. Y varies over the population; a different sample gives different values — **sampling variation**.
3. A model indexed by θ implies a particular θ_0 that is *true*: if your sample were the whole population and the model were true, you would know θ_0.
4. The object of econometrics is to find out θ_0 from the sample.
5. Method: choose an **estimator** or a **test statistic** — a formula — and evaluate it by its behavior over hypothetical repeated samples.

What "good" means
Bias $E(\hat{\theta} - \theta_0)$; variance $E(\hat{\theta} - E\hat{\theta})^2$; MSE; mean absolute deviation; power.

Every one of these is an expectation over samples you did not observe.

Lancaster §A1.1. Note his footnote: "frequentist" refers to this emphasis on repeated-sample behavior.

A1.1 Two features of that approach worth pausing on

- **The population is often imaginary.** In macro, "the population" is a collection of potential periods of economic history — *reruns of history*. Possible to imagine, impossible to do.
- **The sampling distribution is usually hard to calculate.** So frequentists either (a) invoke limit theorems for "large n," or (b) *use a computer to simulate the sampling distribution* — "similarly to the Bayesian use of it to sample from a complicated posterior."

Socratic prompt

Both camps end up at a laptop drawing millions of samples. Same computation, opposite meaning. **What, exactly, is being simulated in each case?** And which of the two simulations could in principle be checked against the world?

A1.2 The Bayesian contrast — points (1)–(2)

(1) The Bayesian must model the data he is about to see
He must formulate $p(y)$, usually via a likelihood $p(y \mid \theta)$ and a prior $p(\theta)$, integrating over θ. But he is *not tied to a population*. A Bayesian macroeconomist can ask himself "what would the economy look like if the parameters took this value or that value?" — that gives a likelihood. No sample, no population required.

(2) Bayes demands more: a complete likelihood
A frequentist can get by with less — e.g. a moment condition $E(g(Y, \theta_0)) = 0$.
"But of course you get what you pay for." A likelihood and a prior imply a posterior and thus inferences that are *exact*. A moment condition yields an estimator whose sampling distribution cannot be deduced — inference must rely on approximations.

Appendix 1, points (1) and (2). Flag point (2): it is the hinge for Part VIII on ABC.

A1.2 The Bayesian contrast — points (3)–(4)

(3) There is only one "estimator"
In a Bayesian procedure the answer is the marginal posterior of the quantity of interest. *The model and the data imply the estimator.* This contrasts sharply with frequentist econometrics "in which, for any model and data, anyone with sufficient ingenuity can come up with a new estimator or test."

(4) Sampling-distribution properties are irrelevant
Bias, MSE, variance, efficiency are features of sampling distributions — hence irrelevant. **"For example, the Gauss–Markov theorem, the foundation stone of traditional econometrics texts, is irrelevant to Bayesian econometrics."**

Socratic prompt
Point (3) is a claim about *sociology* as much as method. What does a literature look like when there is nothing to invent? Is that a feature or a bug?

A1.2 The ambiguity of "repeated sampling": Lancaster's bus example

A well-defined population: Bay Area commuters choosing bus vs. car. Two sampling schemes:

Scheme A. Select exactly 100 bus riders and exactly 100 car drivers.

Scheme B. Toss a fair coin 200 times; heads $\rightarrow$ sample a bus traveler, tails $\rightarrow$ a car traveler.

Scheme B produces, say, **97 bus and 103 car**. Now the frequentist must choose:

- imagine repeated samples of sizes **97 and 103**? or
- imagine repeated samples in which a coin is thrown 200 times and the counts vary?

The distinction matters and the inferences differ. The regression analogue: is the sampling distribution one in which both Y and X vary, or one in which only Y varies?

Appendix 1, point (4). Lancaster: the issue has been discussed by theorists for 75 years, "but never, so far as I am aware, by econometrics texts."

The bus example, Socratically

Socratic prompt

1. Your standard errors changed because of a coin toss that happened *before* you collected data and that has nothing to do with commuting. Defend that.
2. Cox & Hinkley (1974) conclude: condition on 97 and 103; and in regression, hold the regressors at their sample values. That is a *conditioning* argument. Once you start conditioning on what you actually saw, how far can you go before you have become a Bayesian?
3. Lancaster: "It doesn't matter at all to a Bayesian; his inferences are conditional on the data he has, not on what might have happened had he a different sample." Is that a solution, or a refusal to answer the question?

A1.2 Points (5)–(6): operating characteristics and consistency

- **(5)** Some Bayesians *do* report repeated-sampling properties of a Bayes decision — its "operating characteristics" — typically to persuade frequentists. "When these calculations are done the conclusion is almost always that Bayesian procedures have good repeated sampling properties." (Footnote: with a proper prior, Bayes rules are **admissible**.)
- **(6)** Consistency is a dominant interest in traditional econometrics — prizes are won for a new consistent estimator and its asymptotic distribution. But:

Why consistency plays only a limited role
"Consistency refers to the behavior of the sampling distribution when you have *more data than you actually have*! Bayesian inference works from the information that is in your data file."

- The one place limit theorems *do* matter: **MCMC** (Ch. 4), where the "sample" size is the number of draws — chosen by you, unambiguous, and increasable without limit.

Appendix 1, points (5)–(6) and footnotes 5–6.

A1.2 Point (7): the case where the two agree

n iid normal draws, mean μ, variance 1.

Frequentist. $\bar{y}$ is the estimator; over repeated samples $\bar{y} \sim n(\mu, 1/\sqrt{n})$. Interval $\bar{y} \pm 1.96/\sqrt{n}$. Conclusion:

> *"In repeated samples of size n, the interval constructed this way will contain μ on 95% of such occasions."*

Bayesian. With a (improper) uniform prior on μ, the posterior is normal with mean $\bar{y}$, SD $1/\sqrt{n}$. The shortest interval containing 95% of the posterior density is the same interval. Conclusion:

> *"The probability that μ lies between l and u is 0.95."*

Same numbers. Formally different statements. In practice, readers of either take away the same inference.

Socratic prompt

If the numbers coincide so often, why does any of this matter? Give the best case for "it doesn't," then the best case against it.

A1.2 Point (8): what the Bayesian agenda frees up

Much of the econometrics literature develops new estimators/tests or studies the repeated-sampling behavior of existing ones. Lancaster: the Bayesian agenda frees that journal space so the econometrician can focus on **the first four letters of the last noun** — *econ*ometrics:

- What is the relation between my data and my model?
- How do I handle discrepancies between what I **measured** and what theory says I **should have** measured?
- How do I find data more relevant to my theory than the data that exist?
- How do I handle models where $g(y, x) = 0$ has **more than one solution** — as in game theory?

Key idea
The claim is not that Bayes gets better numbers. It is that Bayes moves the hard part of the work from statistics back into economics.

Appendix 1, point (8).

Giocoli overlay I: this argument is 80 years old

Jacob Marschak, 1946, reviewing von Neumann & Morgenstern:

- von Neumann's theory requires that "to be an economic man, one must be a statistical man."
- But in a footnote: Abraham **Wald** was building a statistical decision theory in which "being a statistical man implies being an economic man."

Wald's basic intuition: statistics *is* the science of decision making under uncertainty. A solution to a statistical problem should tell the statistician **what to do**, not just what to say. He called this **inductive behavior**.

Savage, later, sharpened it: statistics must be reinterpreted as a discipline concerned with **behavior rather than assertions** — the *behavioralistic* outlook replacing the *verbalistic* one.

Giocoli (2013), "The Issue" and "The Explorer."

Giocoli overlay I: the paradox we will keep returning to

Giocoli's "little paradox"
Neoclassical economists model their *agents* as Bayesian decision makers — subjective probabilities, Bayes updating, expected utility — while doing their *own* empirical work with frequentist estimation techniques.

"...practitioners follow in their professional routine different behavioral principles than those they impose upon the agents populating their models."

Socratic prompt
Is this a genuine inconsistency, or a division of labor? Note that Lancaster (§1.4.4) will give an answer — and it is not the obvious one.

Appendix 1: the full Socratic set

1. Point (2): Bayes demands a *complete* likelihood, frequentism can survive on a moment condition. Which is the more honest admission of ignorance — specifying a full distribution you half-believe, or specifying only a moment you fully believe?
2. Point (3): "the model and the data imply the estimator." What is lost when ingenuity has nowhere to go?
3. Point (4): Gauss–Markov is declared irrelevant. What else in your first-year sequence goes with it? What survives?
4. Point (6): "consistency refers to the behavior of the sampling distribution when you have more data than you actually have." Is this a fair characterization, or a rhetorical trick?
5. Point (7): when the intervals coincide numerically, does the difference in interpretation have any *decision-relevant* content?
6. Overall: Lancaster says the purpose is "to contrast, perhaps rather starkly." Where do you think he has been unfair to the frequentist?

Part III — Bayes' theorem and the algorithm (§1.3)

§1.3 The theorem itself

For events A, B with $P(B) \neq 0$:

$$P(A \mid B) = \frac{P(B \cap A)}{P(B)} = \frac{P(B \mid A)P(A)}{P(B)}$$

- This is “a universally accepted mathematical proposition.”
- **The disagreement is not about the theorem. It is about its interpretation and its range of application.**

Two questions, taken in order:

1. **How do we interpret $P(\cdot)$?**
2. **To what events may we apply it?**

Lancaster §1.3.

Interpretation: probability as degree of belief

> *"The function $P(\cdot)$ has no interpretation in the mathematical theory of probability; all the theory does is define its properties."*

In this book, $P(A)$ measures **the strength of your belief that A is true**. This is the *subjective* view.

- It is close to ordinary language: "very probable," "highly unlikely."
- It lets an analysis end with: *"in the light of the evidence, theory A is very unlikely whereas theory B is quite probable."*
- **Statements like that are impermissible in traditional econometrics, where theories are only true or false, not more or less probable.**

The one constraint on belief (footnote 5)
Beliefs must obey the laws of probability — they must be **coherent**. Incoherent beliefs expose you to a **Dutch book**: a set of bets you are sure to lose. "Economists in particular are likely to find such a constraint compelling."

Range of application: what may be assigned a probability?

Frequentists restrict probability to events embedded in a repeatable sequence. Lancaster adopts a much more general range:

- the probability that the US was in recession in 1983
- the probability that the moon is made of green cheese
- the probability that the economy is in general equilibrium

“Unless you have the imagination of a Jules Verne there is no sequence of occasions in some of which the economy was in recession in 1983 and in others of which it was not.”

Socratic prompt
Name an economic quantity you care about that *is* embedded in a genuine repeatable sequence. Now name one that is not. Which list was easier? Which list is your dissertation on?

Example 1.1 worked out — two structures

Consumption/income, two structures only: $\theta_1 = (10, 0.9)$ i.e. $C = 10 + 0.9Y$; $\theta_2 = (0, 1)$ i.e. $C = Y$.

Prior: $P(\theta_1) = P(\theta_2) = 0.5$.
Likelihoods of the evidence E:
$P(E \mid \theta_1) = 0.1$, $P(E \mid \theta_2) = 0.6$.
Marginal:

$$P(E) = 0.1(0.5) + 0.6(0.5) = 0.35$$

So E "was not particularly probable."

Posterior:

$$P(\theta_1 \mid E) = \frac{0.05}{0.35} = \tfrac{1}{7} \approx 0.143$$

$$P(\theta_2 \mid E) = \frac{0.30}{0.35} = \tfrac{6}{7} \approx 0.857$$

In odds: prior odds on θ_2 were 1 ("even money"); posterior odds are **6 to 1 on**.

Note what did the work: the **ratio** $0.6/0.1 = 6$. That ratio is the Bayes factor, and it is the whole of the evidence.

Lancaster Example 1.1. Odds on $A = P(A)/(1 - P(A))$.

Example 1.1 in code

```
prior = [0.5, 0.5]                 # P(theta1), P(theta2)
lik   = [0.1, 0.6]                 # P(E|theta1), P(E|theta2)

pE        = sum(lik .* prior)      # 0.35  -- the marginal / predictive
posterior = lik .* prior ./ pE     # [0.1429, 0.8571]

bayes_factor   = lik[2] / lik[1]                # 6.0  -- evidence only
posterior_odds = posterior[2] / posterior[1]    # 6.0 = prior odds (1.0) * BF
```

Key idea
Posterior odds = prior odds × Bayes factor. The data speak only through the likelihood *ratio*. Everything else on this slide is yours, not the data's.

The two exceptions — and they are deep

Evidence changes your beliefs **except** in two cases:

Exception 1: a dogmatic prior
If $P(\theta_1) = 0$ then $P(\theta_1 \mid E) = 0$, whatever E is.
"If you gave no credence at all to a theory you will never learn that it is right."

Exception 2: equal likelihoods
If $P(E \mid \theta_1) = P(E \mid \theta_2)$, the data are equally probable under both, and **"the data carry no information at all about the merits of the two theories."**

These are not edge cases. Exception 1 is why we avoid zero-probability regions (§1.4.2). Exception 2, made to hold *for every possible dataset*, is the definition of **non-identification** (Definition 1.3).

§1.3.1 The distinction that organizes everything

Parameters vs. data
A **parameter** is unknown to you *both before and after* the data are gathered.

Data are unknown before they are gathered and **known afterwards**.

That is the *only* distinction. "Parameter" therefore covers:

- a property of the external world (rotten apples in a barrel);
- an object in a theory (the elasticity of substitution) with **no existence independent of that theory**;
- an index of sub-theories ($\theta_1 =$ theory 1);
- a **function** — if $y = g(x)$ and g is unknown to you, then g is a parameter.

Lancaster §1.3.1: "The material in this section is essential to understanding the point of view taken in this book."

Digression: "random" and "fixed" are not Bayesian words

> *"In the Bayesian approach **all** objects appearing in a model are assigned probability distributions and are random in this sense. The only distinction between objects is whether they will become known for sure when the data are in, in which case they are data(!); or whether they will not become known for sure, in which case they are parameters. Generally, the words 'random' and 'fixed' do not figure in a Bayesian analysis and should be avoided."*

Socratic prompt

Go through the frequentist sentences you have written in the last year: "treat X as fixed," "the random effects model," "fixed effects." Translate each into the parameters/data vocabulary. Which ones survive translation, and which turn out to have been about *sampling schemes* all along?

ALGORITHM 1.1 — Bayes

The algorithm, verbatim

1. Formulate your economic model as a collection of probability distributions conditional on different values for a model parameter $\theta \in \Theta$.
2. Organize your beliefs about θ into a (prior) probability distribution over Θ.
3. Collect the data and insert them into the family of distributions given in step 1.
4. Use Bayes' theorem to calculate your new beliefs about θ.
5. **Criticize your model.**

- Step 1 is the **likelihood**; step 2 the **prior**; step 3 turns a density into a likelihood *function*; step 4 is arithmetic; **step 5 is Chapter 2**.
- "This book could end at this point, though the publisher might object."

Lancaster Algorithm 1.1, §1.3.2.

The five steps, annotated

Step	Object	Where it lives	What can go wrong
1	$p(y \mid \theta)$	§1.4.1	wrong family; non-identified; over-dogmatic
2	$p(\theta)$	§1.4.2	zero probability where truth lives; improper $\to$ improper posterior
3	$\ell(\theta; y^{\text{obs}})$	§1.4.1	conditioning on the wrong thing
4	$p(\theta \mid y)$	§1.4.3	the integral $p(y)$ is intractable $\to$ Part VIII
5	criticism	**Ch. 2**	you skipped it

Key idea

Steps 1–4 are mechanical once you commit. **Step 5 is the only step where the world gets to talk back —** which is why Ch. 2 is co-equal.

The econometric model is likelihood *and* prior

The numerator of Bayes' theorem is the **joint** distribution of data and parameter:

$$p(\theta \mid y) = \frac{p(y, \theta)}{p(y)} = \frac{p(y \mid \theta)\, p(\theta)}{p(y)} \;\propto\; p(y \mid \theta)\, p(\theta)$$

- $p(y, \theta)$ **is your econometric model.**
- $p(y \mid \theta)$: "what you expect to see for every particular value of θ" — your *predictions*.
- $p(\theta)$: your beliefs about which structures are plausible.

Lancaster's definition of a complete model
"From the point of view taken in this book an econometric model is complete only when it specifies **both** the likelihood and the prior."

- $p(y)$ does not involve θ, so for inference *about* θ it can be dropped — hence the $\propto$. **Keep an eye on it anyway: it is the predictive distribution, and it returns in Parts VIII, IX and XI.**

Digression: objectivity, and the seminar as persuasion

> *"Bayesian inference is not 'objective.' Some people, believing that science must be objective and its methods objectively justifiable, find this a devastating criticism. Whatever the merit of this position it does not seem to be the way applied econometrics is practiced."*

Lancaster's description of a typical seminar: the speaker announces her beliefs as a model plus assumptions; **attempts to persuade** the audience that the beliefs are reasonable; promises that the rest will not be contradicted by data; then shows which beliefs survived.

> *"The entire process appears to be subjective and personal. All that a Bayesian can contribute is to ensure that the way in which she revises her beliefs conforms to the laws of probability."*

Socratic prompt

Is this an honest description of the seminars you have sat in? If it is, what is the frequentist apparatus actually doing in those rooms — inference, or credentialing?

Part IV — The likelihood: the information transfer function (§1.4.1)

Two names for one object

$$p(y \mid \theta) \quad \text{vs.} \quad \ell(\theta; y)$$

- As a **density/mass function of** Y at y, given θ: the pdf of Y given θ.
- As a function of θ with the **observed** data y^{obs} plugged in: the **likelihood function**, written $\ell(\theta; y)$ — *semicolon, not conditioning bar*.

Why the semicolon matters

"The likelihood function is not, in general, a probability distribution for θ given data y, nor even proportional to one, though it often is."

It is a function of θ with the data serving as *parameters of that function*. Hence the semicolon.

Key idea

This object is where the data enter — and it is the *only* place they enter. It is the transfer function that carries information from the sample into your beliefs. Prior in, posterior out; the likelihood is the operator.

What choosing a likelihood commits you to

Choosing a likelihood = choosing a **family of probability distributions**, one for each $\theta \in \Theta$. The choice is an art, but it is constrained. It must:

1. **express the economic model** at the center of the investigation;
2. let you determine, from evidence, a distribution over Θ — which values are probable;
3. **allow you to conclude that the model is wrong** (looking ahead to Ch. 2).

Requirement 3 is easy to miss and it is the one with teeth: *a likelihood that cannot be embarrassed by any data is useless.*

Socratic prompt
Write down a likelihood for your current project. Now write down a dataset that would embarrass it. If you cannot, what have you actually assumed?

Example 1.2 — linear regression

Theory: $c = \beta y$ (consumption proportional to income). Deterministic — **inconsistent with any real data**. So embed it: each c_i is a normal variate with mean βy_i and precision τ (precision $= 1/$variance), independent across i:

$$p(c \mid y, \beta) = \prod_i (\tau/2\pi)^{1/2} \exp\big[-(\tau/2)(c_i - \beta y_i)^2\big] \;\propto\; \exp\big[-(\tau/2)\textstyle\sum_i (c_i - \beta y_i)^2\big]$$

Rearranging the sum of squares and dropping terms free of β:

$$\boxed{\ell(\beta; c, y) \;\propto\; \exp\big[-(\tau \textstyle\sum y_i^2/2)(\beta - b)^2\big], \qquad b = \frac{\sum c_i y_i}{\sum y_i^2}}$$

The likelihood has the shape of a normal density, centered at the least squares estimate b, with precision $\tau \sum y_i^2$.

Lancaster Example 1.2 and (1.7). Note the first appearance of least squares — as a *feature of the likelihood*, not as an estimator.

Kernels — the notational habit that makes this bearable

Definition 1.5 — a kernel
A density or mass function typically has the form $k\,g(x)$, where k is the constant that makes it integrate to one. **The remaining portion $g(x)$ is the kernel.**

- Beta kernel: $x^{a-1}(1-x)^{b-1}$. Normal kernel in x: $\exp\{-\tau(x-\mu)^2/2\}$.
- **What counts as the kernel depends on what you think the argument is.** For the normal density viewed as a function of μ *and* τ, the kernel is $\sqrt{\tau}\exp\{-\tau(x-\mu)^2/2\}$.
- "In all Bayesian work and throughout this book we shall systematically retain only the kernels ...Once you get used to it this makes for much easier reading, manipulation, and typing."

This is why Bayes' theorem is usually written $p(\theta \mid y) \propto p(y \mid \theta)\,p(\theta)$.

Example 1.3 — an autoregression

Theory: $y_t = y_{t-1}$. Relax it: $y_t = y_{t-1} + u_t$, $u_t \sim n(0, \tau)$ iid — a **random walk**.

Modeling moves worth naming:

- The theory says nothing about y_1, so condition on it: model $y_2, \ldots, y_T \mid y_1$.
- **Enlarge** the model to $y_t = \rho y_{t-1} + u_t$. The random walk is the special case $\rho = 1$ — "so we can declare the model inconsistent with the evidence if, after having seen the data, $\rho = 1$ seems to be improbable."

$$\ell(\rho; y) \propto \exp\left[-(\tau \textstyle\sum_{t=2}^{T} y_{t-1}^2/2)(\rho - r)^2\right], \qquad r = \frac{\sum y_t y_{t-1}}{\sum y_{t-1}^2}$$

Key idea
Testing a theory means **embedding it in a larger model** in which it is a point, then asking whether that point retains probability. Remember this — it is §2.7, "Enlarging the model."

Example 1.4 — binary choice (probit)

Many important economic variables are binary: you do or do not find work; the economy expands or declines. A binary variate cannot possibly be normally distributed — its sample space has two points.

$$P(Y = 1 \mid x) = p(x), \qquad p(y \mid x) = p(x)^y \big(1 - p(x)\big)^{1-y}$$

Choose $p(x) = \Phi(\beta x)$:

- always in $(0, 1)$, as a probability must be;
- captures "large $x \Rightarrow y$ more likely 1" when $\beta > 0$;
- **allows the idea to be discredited** if β turns out to be negative or zero.

$$\ell(\beta; y, x) = \prod_i \Phi(\beta x_i)^{y_i} \big(1 - \Phi(\beta x_i)\big)^{1-y_i}$$

"No manipulations can simplify this expression further, nonetheless the likelihood can readily be drawn." Note also: $E(Y \mid x) = P(Y = 1 \mid x)$, so **this too is a regression model** — a nonlinear one.

Example 1.5 — Bernoulli trials (the workhorse)

$Y_1, \ldots, Y_n$ iid given θ, $P(Y_i = 1 \mid \theta) = \theta$:

$$p(y \mid \theta) = \theta^s (1-\theta)^{n-s}, \qquad s = \sum_i y_i$$

$$\ell(\theta; y) = \theta^s (1-\theta)^{n-s}, \qquad 0 \le \theta \le 1$$

- This is **the kernel of a beta density** in θ — which is why the beta family will turn out to be the natural conjugate prior (Example 1.7).
- $n = 1$: the likelihood is *linear*, not bell-shaped at all.
- $n = 50$, $s = 1$: concentrated near zero, interior max at $\theta = 0.02$.
- $n = 50$, $s = 25$: looks like a normal curve, symmetric about $1/2$.

Socratic prompt

The $n = 1$ case has a perfectly good likelihood and a perfectly good posterior, but the ML estimate is 1 (or undefined). What does that tell you about which object is primitive?

Simulating and drawing likelihoods

Lancaster's advice: **study the formulae numerically and graphically**, using simulated data where *you* choose the parameters.

```
using Distributions, Random, Statistics
Random.seed!(1)

# Example 1.2: simulate a regression and plot its likelihood in beta
n, beta, tau = 50, 0.9, 1.0
y = rand(Uniform(10, 20), n)
c = rand.(Normal.(beta .* y, 1/sqrt(tau)))

b    = sum(c .* y) / sum(y .^ 2)      # least squares estimate, about 0.899
prec = tau * sum(y .^ 2)              # likelihood precision
grid = range(0.86, 0.94, length = 200)
lik  = pdf.(Normal(b, 1/sqrt(prec)), grid)   # the likelihood, up to a constant
```

"The time series graph showing the actual data seems hard to interpret **but the likelihood is much clearer**."

(Footnote: the likelihood is a *window* onto the data — "but note that many possible windows can be devised.")

Parameters of interest: the same data, a different unknown

Someone runs Bernoulli trials with $\theta = 0.5$ (agreed by both of you) and reports $s = 7$ successes — but **refuses to tell you** n.

Now n is the parameter and θ is data:

$$\ell(n; s, \theta) \;\propto\; \frac{n!}{(n-7)!}\left(\tfrac{1}{2}\right)^n, \qquad n \geq 7$$

- The evidence **constrains the support**: you cannot have had fewer than 7 trials.
- The likelihood points to $n \approx 14 = 7/0.5$ — "most people's guess at the number of trials done to get 7 heads with a fair coin."
- Values above 25 are very improbable.

Key idea
There is no privileged "parameter." Whatever is unknown to you after the data are in is the parameter, and a likelihood can be built for it.

Definition 1.2 — the likelihood principle

Two investigators, same prior, different experiments:

A fixes $n = 20$, observes $s = 7$. Binomial:

$$\ell_A \propto \binom{20}{7}\theta^7(1-\theta)^{13}$$

B samples until 7 successes; the 7th arrives on trial 20. Negative binomial:

$$\ell_B \propto \binom{19}{6}\theta^7(1-\theta)^{13}$$

Same kernel $\Rightarrow$ same posterior $\Rightarrow$ same inference.

Definition 1.2 — The Likelihood Principle
Likelihoods that are proportional should lead to the same inferences (given the same prior).
"What matters for Bayesian inference are the data that **were** observed; the data that might have been seen but were not are irrelevant."

Irrelevance of the stopping rule — press hard here

> *"It doesn't matter whether you chose in advance to do 20 trials or chose to do trials until you observed 7 successes.* ***It doesn't, in fact, matter whether you had anything in your head at all before you began.*** *Maybe you just got bored, or felt ill, after 20 trials."*
>
> *"Most people meeting this implication feel shocked, even outraged, and some continue to do so."*

Lancaster's footnote is the sharp end: frequentist inferences here **differ according to the content of the trialer's head when he stopped**. If you cannot learn that mental state, the frequentist can make no inference at all from 7 successes in 20 trials.

Socratic prompt

1. Would your inference about θ depend on what you knew of the trialer's *intentions*?
2. Would you discard priceless, unrepeatable data because you could not establish the stopping rule?
3. Optional stopping is how clinical trials get abused. If the likelihood principle says stopping rules are irrelevant, how should we think about that abuse?

Giocoli overlay II: the likelihood principle converted Savage

- Savage (1954), *The Foundations of Statistics*: the goal was **Wald's** project — teach statisticians to behave as rational economic agents — using de Finetti's personal probability and the axiomatic method.
- The book's second half tried to reinterpret orthodox inference (point estimation, hypothesis testing, interval estimation) in behaviorist/subjectivist terms **using Wald's minimax rule**.
- It failed. Interval estimation could not be given a behaviorist reading: to a decision maker who must act, "accuracy estimation …could at most satisfy idle curiosity." The book ends abruptly — no summary, no conclusion.
- Savage in 1961: "the minimax theory of statistics is …an acknowledged failure."
- **Savage's own account of what converted him: the discovery of the likelihood principle.**

Key idea
The principle that shocks your students is the same one that turned the most careful frequentist-sympathetic mind of the 1950s into "we radical Bayesians."

Giocoli (2013), "The Strategy," "The Fiasco," and fn. 41.

Populations and samples: a metaphor, not a foundation

Survey sampling has a real population, a real sampling scheme, and a parameter that is a genuine feature of that population. Many applied economists borrow this story to build likelihoods.

Lancaster's assessment:

- It is "a vivid metaphor and helpful to the applied worker" — and it helps persuade an audience, "an important consideration in scientific communication which is, in part, an exercise in persuasion."
- **But:** "economists are rarely, if ever, concerned solely with the properties of some particular, historical population. They wish to generalize, and the basis for generalization is economic theory."
- Footnote 20, with the knife in: *"Economists whose applied work is claimed to rest on 'estimating some feature of a population' seem very rarely to define that population precisely."*

Socratic prompt

Define, precisely, the population your last regression sampled. Take your time.

Identification (Definition 1.3)

Flat spots in the likelihood for **one** dataset: disappointing, not fatal — the prior will distinguish the values.

Flat spots for **every possible** dataset: that is non-identification.

Definition 1.3
θ_1 is **identified** if there is no other θ_2 with $p(y \mid \theta_1) = p(y \mid \theta_2)$ for all $y \in \Omega$. If such a θ_2 exists, the two are **observationally equivalent**.

Example 1.6. n normal draws, mean μ, precision 1, so $\ell \propto \exp\{-(n/2)(\mu - \bar{y})^2\}$. Your theory says $\mu = \alpha + \beta$ (two distinct effects). Then

$$\ell(\alpha, \beta; y) \propto \exp\{-(n/2)(\alpha + \beta - \bar{y})^2\}$$

Every (α, β) with the same sum has identical likelihood — **for every dataset you could ever obtain**.

Why identification is a Bayesian non-event (mostly)

- Flat spots at the top of a likelihood **wreck maximum likelihood**: no unique maximum, singular second-derivative matrices.
- "It is of no special significance from the Bayesian point of view **because Bayesians do not maximize likelihoods — they combine them with priors and integrate them**."
- The qualification: if *all* values on the real line are unidentified and you use an improper flat prior, the posterior is flat too — and that is not allowed.
- Appendix 1, footnote 1, makes the general point: non-identification "is an issue in the construction of **models**, not of inference about them."

Socratic prompt

Historically, non-identification in supply and demand was "a major event in the evolution of our subject." If the Bayesian machine shrugs at it, did we learn something real in 1950, or did we learn a fact about maximum likelihood?

Part V — Exchangeability and de Finetti

The problem with “iid”

It is almost impossible to build an econometric model without writing *“$Y_1, \dots, Y_n$ are independently and identically distributed.”*

But that phrase seems to imply that **somewhere out there is a machine** — like your computer's random number generator — producing a stream of draws from a marginal distribution that exists independently of you.

The subjectivist's objection

"From the subjective point of view, in which probabilities are private and personal, the phrase just cited doesn't look meaningful — **it appears to give probability an objective existence.**"

So some writers, following **de Finetti**, derive their likelihoods from a deeper idea.

Definition 1.4
$Y_1, Y_2, \dots, Y_n$ are **exchangeable** if their joint distribution is unchanged by permutation of the subscripts: e.g. for $n = 3$, $p(Y_1, Y_2, Y_3) = p(Y_2, Y_3, Y_1)$, and so on for all six permutations.

- Exchangeability implies equal means, equal variances, and **the same correlation for every pair**.
- **Exchangeable sequences need not be iid. iid sequences are always exchangeable.**
- Whether a sequence is exchangeable **is a matter of judgement** — *your* judgement: *"You form a judgement about $p(Y_1, Y_2, Y_3)$ and then you are asked to form a judgement about $p(Y_2, Y_3, Y_1)$: would you give a different answer? If you would not …then your beliefs about Y_1, Y_2, Y_3 are exchangeable."*

Key idea
Note the grammar: it is not the coin that is exchangeable. **Your beliefs** are exchangeable.

de Finetti's representation theorem

If a sequence of binary random variables is exchangeable **for every** n, then

$$p(y_1, y_2, \dots, y_n) = \int_0^1 \theta^s (1-\theta)^{n-s}\, dF(\theta), \qquad s = \sum y_i$$

Read the right-hand side carefully

That is **exactly** a Bayesian marginal data distribution: a likelihood $\theta^s(1-\theta)^{n-s}$ integrated against a prior distribution function $F(\theta)$.

> "***So exchangeability implies the existence of a likelihood and a prior.*** *It is an amazingly powerful idea. It means that you have no need to start your modelling with the assertion that a collection of random variables are independent and identically distributed. You can instead merely state that your beliefs about them are exchangeable and this will automatically imply that the model takes the form of a likelihood and a prior.*"

What the theorem actually buys you

1. **The prior is not an optional extra you bolted on.** Judge your data exchangeable and a prior *falls out of the judgement* whether you like it or not.
2. θ **is derived, not assumed.** In the representation, θ appears as a mathematical consequence of a symmetry in your beliefs — it is the limiting relative frequency $\lim s/n$, a *feature of your predictive distribution*, not a physical constant.
3. **The primitive object is $p(y_1, \ldots, y_n)$ — the predictive distribution over observables.** Parameters are a device for representing it. (Part IX will lean on this.)

Lancaster's honest coda: *"Having said this, it is the case in almost all practice by Bayesian econometricians that they begin modeling in the conventional way without any deeper justification. We shall mostly follow that path in this book."*

Exchangeability, Socratically

Socratic prompt

1. Are daily returns on SPY exchangeable in your beliefs? If a 2008 observation and a 2017 observation were swapped, would your joint distribution change? (If yes — and it should be yes — what have you just committed yourself to modeling?)
2. Are the 50 states exchangeable in a panel regression? Before you see the data? After?
3. Lancaster says the judgement is yours. Two economists disagree about exchangeability of the same sequence. What kind of disagreement is that — empirical, theoretical, or aesthetic?
4. If exchangeability *implies* a prior, what happens to the objection "I don't want to use a prior"?

Part VI — The prior (§1.4.2)

First: you don't have to believe it

*"Although $p(\theta)$ represents your beliefs **you don't need to believe it!**"*

You may — and should — examine the impact of alternative beliefs on your posterior. Ask "what if?" This is **sensitivity analysis**, and it applies to the likelihood just as much as to the prior.

Second permission, more surprising:

*"Although we usually interpret Bayes' theorem as operating in temporal order, prior $\rightarrow$ data $\rightarrow$ posterior, **this is not a necessary interpretation** and it is formally quite legitimate to allow your 'prior' beliefs to be influenced by inspection of the data."*

Sherlock Holmes: *"It is a capital mistake to theorize before one has data."*

Why this is legal: Bayes' theorem does not restrict the choice of prior. It only prescribes how beliefs *change*.

Encompassing (vague) priors

You are writing for an audience. A prior that conflicts sharply with your readers' beliefs makes your work uninteresting to them:

> *"You will be saying 'If you believed A before seeing the data you should now believe B.' But this will be met with 'So what, I don't believe A.'"*

So for public scientific work use priors **not sharply inconsistent with any reasonable belief**.

- Uniform θ on $[0, 1]$ for a Bernoulli parameter *represents* nobody's belief — someone convinced $\theta \in (0.4, 0.6)$ does not hold it — but it is **not inconsistent with** that belief.
- Such a prior **encompasses** reasonable beliefs. These are called **vague**.
- **Never assign zero probability to a region.** Zero times anything is zero: you can never learn that the region is where the truth lives. (Exception 1 from Part III.)
- But: any model contains dogmatic assertions, "since without them the theory would be vacuous." So the advice can never be strictly fulfilled.

Natural conjugacy (Example 1.7)

Definition
A prior is **natural conjugate** if, after multiplication by the likelihood, the posterior lies in the same family. Only the *parameters* change with data — not the mathematical form.

Bernoulli. Likelihood kernel $\theta^s(1-\theta)^{n-s}$. Any prior $\propto \theta^{a-1}(1-\theta)^{b-1}$ gives a posterior of the same form. So the conjugate family is the **beta family**.

$$\text{Beta}(a,b) \ \times \ \theta^s(1-\theta)^{n-s} \ \longrightarrow \ \text{Beta}(a+s,\ b+n-s)$$

- A conjugate prior "can be interpreted as a posterior arising from some earlier, possibly fictional, evidence" — $a-1$ prior successes and $b-1$ prior failures.
- **Honest caveat from Lancaster:** "The property of natural conjugacy was of more importance in the days when posterior distributions were computed analytically and not, as now, numerically."

Key idea
Conjugacy is a convenience, not a principle. Part VIII is about life without it.

Improper priors — and why a proper posterior can still emerge

$p(\theta) \propto 1$ on $(-\infty, \infty)$ does not integrate. It is **not a probability distribution**. Yet it is used constantly. Why?

Example 1.8. n normal draws, mean θ, precision 1: $\ell(\theta; y) \propto \exp\{-(n/2)(\theta - \bar{y})^2\}$ — a perfectly proper normal kernel with mean $\bar{y}$ and precision n, for any $n > 0$. Multiply by the improper flat prior: **the posterior is proper.**

Example 1.9 (the limit argument). Take a proper $n(0, \tau)$ prior. Posterior is normal with

$$\text{mean } \frac{n\bar{y}}{n+\tau}, \qquad \text{precision } n+\tau$$

Let $\tau \to 0$: mean $\to \bar{y}$, precision $\to n$ — exactly the flat-prior answer.

> *"A proper prior with τ sufficiently small will produce much the same posterior as an improper, uniform, prior."*

Warning (footnote 26): in other models improper priors lead to improper posteriors.

The principle of precise measurement

Look again at the likelihood plots: **the likelihood is effectively zero over almost all of the parameter space.**

Therefore the prior is multiplied by (almost) zero almost everywhere — and **what you believed in those regions cannot matter.**

The practical consequence
A prior intended to be roughly neutral need only be neutral **where the likelihood is non-negligible**. How it behaves elsewhere is of no consequence.

So a formally-uniform-on-the-real-line prior is practically equivalent to one that is uniform where the likelihood lives and does anything bounded outside it.

Corollary: "Many of the standard calculations in econometrics — least squares regression and all maximum likelihood methods — **can be thought of as flat prior Bayes**."

Socratic prompt
If ML is flat-prior Bayes, what exactly were you objecting to when you objected to priors?

Jeffreys' invariant priors

The problem. We can parametrize the same model in infinitely many ways: σ, σ^2, $\tau = 1/\sigma^2$. A rule that gives a prior for σ and the *same rule* applied to σ^2 should yield **the same posterior beliefs about the same quantity**.

Jeffreys: *"equivalent propositions should have the same probability."*

Jeffreys' rule

$$p(\theta) \;\propto\; I_\theta^{1/2}, \qquad I_\theta = -E\left(\frac{\partial^2 \log \ell(\theta; y)}{\partial \theta^2}\right)$$

where the expectation is over $p(y \mid \theta)$.

Invariance follows from $I_\psi = I_\theta (\partial\theta/\partial\psi)^2$, hence $I_\psi^{1/2} = I_\theta^{1/2} |\partial\theta/\partial\psi|$ — exactly the Jacobian needed for the change of variable to go through.

Jeffreys, examples 1.10 and 1.11

Example 1.10 — normal scale.
Working in σ: $I_\sigma = 2n/\sigma^2 \Rightarrow p(\sigma) \propto 1/\sigma$.
Working in τ: $I_\tau = n/2\tau^2 \Rightarrow p(\tau) \propto 1/\tau$.
Change variables from τ back to σ: **you get the identical posterior.** Invariance confirmed.

Example 1.11 — Bernoulli.

$$I_\theta = \frac{n}{\theta(1-\theta)} \Rightarrow p(\theta) \propto \frac{1}{\sqrt{\theta(1-\theta)}}$$

This is Beta$(1/2, 1/2)$: **proper, but U-shaped.** Under it the *least* likely value of θ is $1/2$.
"Notice that Jeffreys' prior is not uniform, as one might, perhaps, have anticipated."

Socratic prompt
Jeffreys' rule takes an expectation over y — a **repeated sampling calculation**. Lancaster notes this "certainly violates the likelihood principle." Can an objective Bayesian prior be built without smuggling frequentism back in?

Hierarchical priors (Example 1.12)

When $\theta = (\theta_1, \ldots, \theta_n)$ has elements that are **similar** — same units, same role in the model (one coefficient per agent, one precision per agent) — express that similarity by making them draws from a **parent** distribution:

$$p(\theta) = \int p(\theta \mid \lambda)\, p(\lambda)\, d\lambda$$

Example 1.12. Precisions τ_i from a gamma parent $\lambda = (\alpha, \beta)$:

$$p(\tau \mid \lambda) \propto \prod_i \tau_i^{\alpha-1} e^{-\beta\tau_i}, \qquad p(\tau, \lambda \mid y) \propto \prod_i \tau_i^{\alpha-1/2} \exp\{-\tau_i(\beta + y_i^2/2)\}\, p(\lambda)$$

This is the basis of **robust** Bayesian analysis: it relaxes the dogmatic restriction that all observations share one precision.

Hierarchies dissolve the likelihood/prior boundary

Same model, two presentations:

$$\underbrace{\ell(\theta; y)}_{\text{likelihood}} \; \underbrace{p(\theta \mid \psi) p(\psi)}_{\text{prior}} \qquad \text{vs.} \qquad \underbrace{\ell(\psi; y)}_{\text{likelihood}} \; \underbrace{p(\psi)}_{\text{prior}}, \quad \ell(\psi; y) = \int \ell(\theta; y) \, p(\theta \mid \psi) \, d\theta$$

Which is the likelihood?

"The answer is that **it doesn't really matter**; all that does matter is the product of prior and likelihood, which can be taken as $p(y, \theta, \psi)$ or as the y, ψ marginal $p(y, \psi) = \int p(y, \theta, \psi) \, d\theta$."

Socratic prompt

Appendix 1 point (2) said Bayes is "more demanding" because it requires a full likelihood, and the sharp line between likelihood and prior is what most frequentist objections attack. If the line is *arbitrary*, what is left of the objection — and what is left of the demand?

Parameter separation and information orthogonality (Example 1.13)

Default priors are well understood for **scalar** parameters and much less so for vectors. One strategy: reparametrize so the problem *becomes* many scalar problems.

Regression $c_i = \alpha + \beta y_i + \varepsilon_i$: the likelihood does not factor because $X'X$ is not diagonal ($\sum y_i \neq 0$). **But we can make it true.** Write

$$c_i = \underbrace{(\alpha + \beta\bar{y})}_{\alpha^*} + \beta(y_i - \bar{y}) + \varepsilon_i$$

Now $X'X = \text{diag}\big(n,\ \sum(y_i - \bar{y})^2\big)$ and the likelihood **separates**: one normal curve centered at mean consumption, one centered at the least squares slope.

Key idea
Multiplicatively separable likelihood $\Rightarrow$ additively separable log likelihood $\Rightarrow$ cross-partials vanish $\Rightarrow$ **diagonal information matrix**. Searching for $g(\theta)$ that diagonalizes I is searching for **information-orthogonal** parametrizations — and those simplify default priors in high dimensions.

Part VII — The posterior (§1.4.3)

Example 1.14 — the beta-binomial update

Prior Beta(a, b), likelihood $\theta^s(1-\theta)^{n-s}$:

$$p(\theta \mid y) \propto \theta^{s+a-1}(1-\theta)^{n-s+b-1} = \text{Beta}(s+a,\ n-s+b)$$

$$E(\theta \mid y) = \frac{s+a}{n+a+b}, \qquad V(\theta \mid y) = \frac{(s+a)(n-s+b)}{(n+a+b)^2(n+a+b+1)}$$

If s and n are both large with $s/n = r$: $E(\theta \mid y) \approx r$, $V(\theta \mid y) \approx r(1-r)/n$.

The moral
"As evidence accumulates, the posterior becomes **dominated by the likelihood**, virtually independent of the shape of the prior, and ultimately converges to a point."

Calculation 1.5 (the one-trial case). One success in one trial, uniform prior: posterior is $p(\theta \mid y) = 2\theta$, and $E(\theta \mid y) = 2/3$. The ML estimate "either doesn't exist or is 1."

Reporting the posterior: four ways

1. **Draw it**
For scalar θ — or a scalar function of a vector θ, "as it often is in econometrics" (e.g. $\partial P(Y=1 \mid x)/\partial x_j = \beta_j \phi(x\beta)$) — **the best way to convey the posterior is to plot it**.

2. **Report its moments**
Posterior mean (or median) and posterior standard deviation. This is the object that *looks* like the traditional table — but the SD is the SD of your beliefs, not of a sampling distribution.

3. **Report a highest posterior density (HPD) region**
An interval containing 95% of posterior probability such that **no point inside has lower density than any point outside**. Interpretation: "the probability that ρ lies in $[0.939, 1.084]$, given the model and data, is 0.95." *It's as simple as that.*

4. **Calculate the marginals**
$p(\theta_3 \mid y) = \iiint p(\theta \mid y)\, d\theta_1 d\theta_2 d\theta_4 \cdots$ — a $k-1$ dimensional integration. **"This is, in general, a hard problem."**

HPD vs. confidence interval — say it precisely

Confidence interval (the arcane one)

"If you calculated your interval in the same way over many hypothetical repeated samples of the same size and using the same model, then 95% of such intervals would contain within them the 'true' value of θ."

A statement about **the procedure**.

HPD interval

"The probability that ρ lies within 0.939 to 1.084, given the model and data, is 0.95."

A statement about **the parameter**, conditional on what you saw.

Lancaster's AR(1) worked numbers: $r = 1.011$, $s = 0.037$; 95% HPD $= [0.939, 1.084]$; 99% $= [0.916, 1.107]$.

Socratic prompt

Note that $\rho = 1$ sits inside both intervals. A frequentist unit-root test and this HPD would "accept." Are they saying the same thing? (Hold this — it is a famous Bayesian/frequentist battleground in time series.)

Theorem 1.1 — the large sample normal approximation

Theorem 1.1
For sufficiently large n, θ is approximately normal with mean equal to the **posterior mode** $\hat{\theta}$ and precision equal to $-H(\hat{\theta})$, where H is the hessian of the *log posterior*.

Under a uniform prior the posterior equals the likelihood, so $-H$ is the negative second derivative of the log likelihood at the mode — the **observed information**.

Three warnings Lancaster attaches, all worth reading aloud:

1. The approximation holds for **any** parametrization — but it can be accurate for θ and badly wrong for $g(\theta)$, and vice versa.
2. Multivariate normals are unimodal, so the theorem is useless for **multimodal** posteriors.
3. **"It is almost always not the sample size that determines when the sample is large"** but some other function of the data, e.g. $\sum(x_i - \bar{x})^2$. That can be tiny with huge n. (Ch. 8: 36,000 observations with seven parameters is "a very small sample indeed.")

Likelihood dominance and Theorem 1.2

$$\log p(\theta \mid y_1, \dots, y_n) = \log \ell(\theta; y_1, \dots, y_n) + \log p(\theta)$$

As data accumulate the **likelihood term grows in modulus while the prior term stays fixed**. So in large samples the likelihood dominates — *provided* $p(\theta) > 0$ where the likelihood concentrates.

Example 1.16. Bernoulli, three very different beta priors (uniform; Beta$(3,3)$; Jeffreys Beta$(1/2, 1/2)$). At $n = 5$ they still differ; at $n = 20$ they are nearly indistinguishable.

Theorem 1.2 — convergence

With a discrete Θ, iid data, and an identification condition stated as a **Kullback–Leibler** divergence $\int p(x \mid \theta_t) \log[p(x \mid \theta_t)/p(x \mid \theta_s)] dx < \infty$ (and > 0) for all $s \neq t$: $\lim_n p(\theta_t \mid x) = 1$.

> *"This theorem forms a precise statement of how individuals with quite different initial beliefs can be brought into ultimate agreement by the accumulation of evidence." But note: they must agree on the likelihood, even while disagreeing in their priors.*

Sampling the posterior — the hinge of the whole book

"The material in this section is essential to understanding the point of view taken in this book."

Draw `nrep` realizations of θ from the posterior. The output is a matrix: one row per draw, one column per element of θ.

$$\begin{bmatrix} \theta_1^{(1)} & \theta_2^{(1)} \\ \theta_1^{(2)} & \theta_2^{(2)} \\ \vdots & \vdots \\ \theta_1^{(\mathsf{nrep})} & \theta_2^{(\mathsf{nrep})} \end{bmatrix}$$

- The whole matrix is a sample from the **joint** posterior.
- The jth column is a sample from the **marginal** posterior of θ_j.
- **"To study the distribution of θ_1 given the data, just ignore the second column. It's as simple as that."**
- For any function: apply $g(\cdot)$ **row-wise** and you have a sample from the posterior of $g(\theta)$.

Why sampling replaces mathematics

- A law of large numbers applies (draws need not even be independent), so moments of $g(\theta)$ from `nrep` draws converge to the true posterior moments.
- **You choose `nrep`.** So you can make any feature of the posterior — mean, precision, density, quantiles — as accurate as you like.

  *"It follows that **if you can sample the distribution in question you can know it with arbitrary accuracy.** ...Increasingly, difficult mathematics is being abandoned in favor of computer power."*

Key idea

This is why the intractable integral in $p(y)$ and in the marginalization (1.35) stopped being fatal around 1990. It is also the *entire* justification for Part VIII.

Lancaster §1.4.3, "Sampling the posterior," and footnote 38.

Example 1.17 — the pattern to internalize

Probit with $\theta = (\beta_0, \beta_1, \beta_2)$. The quantity of economic interest is a **scalar function** of all three:

$$\gamma = \frac{\partial P(Y = 1 \mid x)}{\partial x_1} = \beta_1 \phi(x\beta)$$

Finding $p(\gamma \mid y)$ analytically requires a two-dimensional integration. The modern way:

1. Sample the joint posterior of $(\beta_0, \beta_1, \beta_2)$ — Lancaster uses BUGS, 10,000 draws.
2. For **each** draw, compute γ at the x of interest.
3. You now have 10,000 draws from $p(\gamma \mid y)$. Plot it. Read off whatever you need.

Lancaster's result: β_2 posterior centered at -0.21 (ML estimate: also -0.21; the data-generating value was -0.5, "one of the less probable values, though it would still lie within a 95% hpd region"). The effect γ is "certainly positive and most probably about 0.17."

The same thing, in Julia

```julia
using Distributions, Statistics

# `chain` holds nrep draws of (b0, b1, b2) from the posterior
# (conjugate sampling, MCMC, ABC -- the source does not matter here)

g = [b[2] * pdf(Normal(), b[1] + b[2]*x1 + b[3]*x2) for b in eachrow(chain)]

mean(g)                          # posterior mean of the marginal effect
quantile(g, [0.025, 0.975])     # 95% credible interval
mean(g .> 0)                     # P(effect is positive | data)  <- ask this directly
```

Socratic prompt

That last line — `mean(g .> 0)` — is the posterior probability that the effect is positive. Write down the frequentist object that answers the same question.

Take your time. That is the point.

Part VIII — No conjugate prior: Monte Carlo and ABC

Where the difficulty actually is

Two integrals stand between you and an answer:

$$\textbf{(a)} \quad p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta \qquad\qquad \textbf{(b)} \quad p(\theta_j \mid y) = \int p(\theta \mid y)\, d\theta_{-j}$$

- **(a)** normalizes the posterior. Conjugacy makes it disappear because you *recognize the kernel* and never do the integral.
- **(b)** is the $k - 1$ dimensional marginalization of §1.4.3 — "in general, a hard problem."

Two solutions, in Lancaster's words

(i) Approximations to the posterior — "fairly old and of wide though not universal applicability" (Theorem 1.1).

(ii) Computer-assisted sampling — "new, rather easy and of what is apparently universal application."

The trouble with (i): "the only way of finding out the accuracy of an approximation to your posterior is to calculate the exact distribution — which is what you wanted to avoid doing!"

Monte Carlo is just integration by sampling

For any g,

$$E[g(\theta) \mid y] = \int g(\theta)\, p(\theta \mid y)\, d\theta \;\approx\; \frac{1}{M} \sum_{m=1}^{M} g(\theta^{(m)}), \qquad \theta^{(m)} \sim p(\theta \mid y)$$

Every posterior summary is an expectation of *some* g:

Want	$g(\theta)$
posterior mean	θ
posterior variance	$(\theta - \bar{\theta})^2$
$P(\theta > 0 \mid y)$	$\mathbf{1}\{\theta > 0\}$
marginal effect	$\beta_j \phi(x\beta)$
value at risk	a quantile — *sort the draws*

Key idea

You never needed a formula for the posterior. You needed **draws** from it. That is the whole conceptual shift of the last thirty years.

The ladder of samplers

1. **Direct sampling.** Conjugate case: `rand(Beta(a+s, b+n-s), M)`. Done.
2. **Grid / inversion.** Scalar θ: evaluate the unnormalized posterior on a grid, normalize numerically. Perfectly respectable in one or two dimensions; dies in higher ones.
3. **Rejection sampling.** Propose from q, accept with probability $\propto \tilde{p}(\theta)/(Mq(\theta))$. Exact, but needs an envelope.
4. **Importance sampling.** Draw from q, weight by $\tilde{p}(\theta)/q(\theta)$. No rejection, but weights can degenerate.
5. **MCMC** (Metropolis–Hastings, Gibbs, HMC/NUTS). Build a Markov chain whose *stationary* distribution is the posterior. Draws are dependent — **which is fine**: "it is desirable, but not essential, that the rows of your output matrix be independent." $\rightarrow$ **Lancaster Chapter 4**; BUGS then, Turing.jl / Stan now.
6. **ABC.** For when you cannot even *evaluate* the likelihood.

Note where the ladder is climbing: away from *knowing* $p(y \mid \theta)$ analytically, toward only being able to *use* it.

Approximate Bayesian Computation: the idea

The situation
You can **simulate** data from your model — for any θ, you can generate a plausible y — but you cannot write down or evaluate $p(y \mid \theta)$.

Recall Appendix 1, point (2): Bayes is "more demanding" precisely because it "requires a likelihood, a complete statement of what values of Y are probable ...for each potential value of θ."

ABC pays a different price. It requires a *simulator*, not a formula.

Where this happens in economics and finance:

- structural / agent-based macro models with no closed-form implied density
- equilibrium models solved numerically (DSGE with occasionally-binding constraints)
- microstructure and queueing models of order flow
- any model you can only run **forward**

ABC is the Bayesian algorithm, run forward

ALGORITHM (rejection ABC)
For $m = 1, \dots, M$:

1. Draw $\theta^* \sim p(\theta)$ *(step 2 of Algorithm 1.1)*
2. Simulate $y^* \sim p(y \mid \theta^*)$ *(step 1 of Algorithm 1.1)*
3. Accept θ^* if $d\big(S(y^*),\, S(y^{\text{obs}})\big) \le \varepsilon$

The retained draws are a sample from $p(\theta \mid d(S(y), S(y^{\text{obs}})) \le \varepsilon)$.

Three choices, three costs:

- $S(\cdot)$**, the summary statistic.** If S is **sufficient**, no information is lost. Otherwise you are conditioning on less than your data.
- ε**, the tolerance.** $\varepsilon \to 0$ gives the exact posterior and an acceptance rate $\to 0$. This is the entire tradeoff.
- $d(\cdot,\cdot)$**, the metric**, and the scaling of the components of S.

ABC in Julia: exact answer, no likelihood used

```julia
using Distributions, Random, Statistics
Random.seed!(20260813)

n, s_obs, M = 20, 7, 400_000       # 20 Bernoulli trials, 7 successes

draws = Float64[]
for m in 1:M
    ts = rand(Uniform(0, 1))       # 1. draw theta* from the prior
    ys = rand(Binomial(n, ts))     # 2. SIMULATE data (no pdf evaluated!)
    if ys == s_obs                 # 3. accept: S = sum(y), eps = 0
        push!(draws, ts)
    end
end

mean(draws), std(draws)            # 0.3636, 0.1003
```

ABC gives $0.3636,\ 0.1003$; the conjugate answer `Beta(1+7, 1+13)` gives $0.3636,\ 0.1003$. **We never wrote down** $\theta^s(1-\theta)^{n-s}$ — we only ran the model forward.

Verified: ABC mean 0.3626, sd 0.1002 vs. exact Beta(8,14) 0.3636, 0.1003; KS $p = 0.54$.

Why that worked exactly — and the beautiful side effect

Why exact: $s = \sum y_i$ is **sufficient** for θ in the Bernoulli model, and $\varepsilon = 0$, so accepting on $\{y^* = s_{\text{obs}}\}$ conditions on *exactly* the information in the data. ABC here is not approximate at all.

The side effect: you just estimated $p(y)$
The **acceptance rate** is a Monte Carlo estimate of

$$P(S = s_{\text{obs}}) = \int \binom{n}{s} \theta^s (1-\theta)^{n-s} p(\theta)\, d\theta$$

which is **the prior predictive —** the denominator of Bayes' theorem that we "dropped."

With a uniform prior this is exactly $1/(n+1) = 1/21 = 0.0476$. Simulation gave 0.0472.

Key idea
The quantity you discard as a constant in Ch. 1 is the quantity Ch. 2 uses to *check the model* (§2.4.2, the prior predictive / marginal likelihood) and *choose* between models (§2.6, posterior odds). It was never really a nuisance.

The honest limitations of ABC

1. **Sufficient statistics rarely exist** outside the exponential family. In practice S is a hand-picked vector of moments, autocorrelations, quantiles — and you are then doing exact inference for a *different, coarser* problem: $p(\theta \mid S(y))$, not $p(\theta \mid y)$.
2. $\varepsilon > 0$ **inflates the posterior.** The ABC posterior is always at least as diffuse as the true one. Reporting an ABC posterior without reporting ε is like reporting an estimate without a standard error.
3. **The curse of dimension.** Acceptance rates collapse as $\dim(S)$ grows. Rejection ABC is a teaching device; practice uses ABC-MCMC, ABC-SMC, or regression adjustment.
4. **It is not free of the likelihood — it is free of *evaluating* it.** The simulator *is* a likelihood; you have just declined to write it down.

Socratic prompt

Does ABC obey the likelihood principle?

(Careful. With sufficient S and $\varepsilon = 0$: yes. With a chosen summary and $\varepsilon > 0$: you are conditioning on *a function of* the data you chose in advance — which starts to look uncomfortably like choosing a test statistic.)

Where MCMC fits (a pointer to Chapter 4)

```
using Turing

@model function bernoulli_model(y)
    theta ~ Beta(1, 1)                  # step 2: the prior
    for i in eachindex(y)
        y[i] ~ Bernoulli(theta)         # step 1: the likelihood
    end
end

chain = sample(bernoulli_model(y), NUTS(), 4_000)
```

- Lancaster's BUGS program (Example 1.17), twenty years on. **Structure identical: tell it your likelihood, tell it your prior.**
- Appendix 1, fn. 6: MCMC is the *one* place limit theorems matter to a Bayesian — here the "sample" size "is entirely under the control of the investigator."

Socratic prompt

Convergence diagnostics, burn-in, effective sample size — statistical questions or numerical-analysis questions? Does it matter for how honestly we report them?

Part IX — Posterior prediction (§1.3.2, §2.4–2.5)

Prediction was in Bayes' theorem the whole time

The denominator we kept dropping:

$$p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta$$

Lancaster names it twice: **the marginal distribution of the data** and **the predictive distribution of the data**. Chapter 2 calls it the **prior predictive** (or *marginal likelihood*).

From §1.5, the summary of the whole chapter:

> "*You must formulate your theory as a conditional probability statement for the data you are about to see and a prior over the parameters.* ***This is equivalent to making a simple statement about what you think the data should look like*** *on your theory, since* $\int p(y \mid \theta) p(\theta) d\theta = p(y)$*, a probability distribution for the data.*"

Key idea
Specifying a likelihood and a prior *is* making a prediction about the data. Model criticism (Ch. 2) is nothing more than comparing that prediction to what showed up.

The two predictive distributions

Prior predictive — §2.4.2

$$p(y) = \int p(y \mid \theta)\, p(\theta)\, d\theta$$

"Indicates what your data should look like, given your model, **before you have seen them**." Also called the marginal likelihood. Basis of formal model checks.

Posterior predictive — §2.5, equation (2.9)

$$p(\tilde{y} \mid y^{\mathsf{obs}}) = \int p(\tilde{y} \mid y^{\mathsf{obs}}, \theta)\, p(\theta \mid y^{\mathsf{obs}})\, d\theta$$

"Indicates what **another realization** from your model should look like given the data you have seen." A future observation, a past one, or a replication of your dataset.

"The integrand is the joint density of $\tilde{y}$ and θ and the only difference from (2.4) is that we have now included y^{obs} in the conditioning set. **This is how Bayesians predict.**"

ALGORITHM 2.5 — Bayesian prediction

Verbatim

1. Sample θ from its posterior distribution.
2. Insert that realization into the conditional distribution given θ (and y^{obs} if necessary) and sample $\tilde{y}$ from it.
3. Repeat `nrep` times — you now have that many realizations from $p(\tilde{y} \mid y)$.

```
# generic two-line predictive simulation
theta_draws = rand(posterior, nrep)                                   # step 1
ytilde      = [rand(observation_model(t)) for t in theta_draws]  # step 2
```

Note the shape of it: **one draw of θ per draw of $\tilde{y}$.** Parameter uncertainty and outcome uncertainty are compounded, not stacked separately.

The plug-in fallacy — Lancaster's digression

What frequentist practice does

"Frequentist practice is to replace $p(\tilde{y} \mid y, \theta)$ by $p(\tilde{y} \mid y, \hat{\theta})$ and prediction is based on this object. This is called a **plug-in** method.

It has no theoretical justification except as an approximation to (2.9) when θ is concentrated about $\hat{\theta}$."

Unpack the failure mode:

$$\underbrace{p(\tilde{y} \mid y, \hat{\theta})}_{\text{one } \theta\text{, no parameter uncertainty}} \qquad \text{vs.} \qquad \underbrace{\int p(\tilde{y} \mid y, \theta)\, p(\theta \mid y)\, d\theta}_{\text{averaged over your uncertainty}}$$

- Plug-in is **always** too narrow. It reports the world's variability and silently drops yours.
- The gap closes as the posterior concentrates — and is worst exactly when you have little data, a long horizon, or a nonlinear function of θ.
- **In Part XI this stops being philosophy and starts being a number on a risk report.**

Example 2.11 — posterior prediction of an autoregression

Model: $y_t = \rho y_{t-1} + u_t$, $u_t \sim n(0,1)$, flat prior. Posterior: $\rho \mid y \sim n(r,\ \sum_t y_{t-1}^2)$, with r the least squares estimate.

Predicting the next observation $\tilde{y} = y_{n+1}$:

$$p(\tilde{y} \mid y) \propto \int \exp\left[-\tfrac{1}{2}(\tilde{y} - \rho y_n)^2\right] \exp\left[-\tfrac{s^2}{2}(\rho - r)^2\right] d\rho$$

$$\implies \quad \tilde{y} \mid y \ \sim \ n\left(r y_n,\quad \text{precision} = \frac{\sum_t y_{t-1}^2}{y_n^2 + \sum_t y_{t-1}^2}\right)$$

Read the precision

"Notice that **the precision of your prediction is always less than one**, even though the precision of the variation about your model, τ, is one."

The shortfall is exactly the parameter uncertainty in ρ, and it is **larger when y_n is large** — forecasting from an extreme point is harder, and the formula knows it. The plug-in forecast reports precision 1 regardless.

Exchangeability, one more time — prediction is the primitive

de Finetti's theorem said: exchangeable beliefs $\Rightarrow$ your joint distribution over observables has the form $\int \prod p(y_i \mid \theta)\, dF(\theta)$.

Turn it around
The thing you actually have beliefs about is $p(y_1, \dots, y_n, \tilde{y})$ — **observables**. θ is the device that represents those beliefs compactly.

On that reading, $p(\tilde{y} \mid y)$ is not a derived quantity at all. **It is the object of the exercise**, and the posterior over θ is scaffolding.

Socratic prompt

1. Which would you rather be asked to defend in a seminar: your posterior for β, or your predictive distribution for next quarter's data?
2. Which one can be *checked*?
3. If only the second can be checked, why do our tables report the first?

Part X — Decisions (§1.4.4): econometrics vs. decision theory

§1.4.4 — Lancaster's position, stated plainly

> *"Many writers prefer to view the problem of inference from data as that of* ***making a decision or taking an action****. ...This is a point of view that is attractive to economists, who want to consider the agents whose behavior they model to be rationally coping with the uncertainties they face."*

And then he declines

"**This book does not take a decision theoretic perspective**, though it is not inconsistent with one. This is because the problem faced by most economists does not seem well described as one of decision. **It seems more like that of sensibly and concisely reporting their findings**, and for this the recommended procedure is to draw the marginal(s) of the object of interest. **This leaves it up to others — for example policy makers — to use your report as a basis for decision making.**"

That is a substantive methodological choice, not a throwaway. It deserves a fight.

The decision-theoretic apparatus, briefly

You have a model, you have seen y, you hold $p(\theta \mid y)$, and you must produce a single number $d = d(y)$.

Bayes decision
Given a **loss function** $L(d, \theta)$ — the loss of deciding d when the parameter is θ — the Bayes decision minimizes **expected posterior loss**:

$$\hat{d} = \arg\min_{d \in \Theta} \int_{\Theta} L(d, \theta)\, p(\theta \mid y)\, d\theta$$

Example 1.18 — squared error loss. $L(d, \theta) = (d - \theta)^2 \Rightarrow \hat{d} = E(\theta \mid y)$, the posterior mean. (Differentiate under the integral.)

Other losses, other reports: absolute error $\rightarrow$ posterior **median**; 0–1 loss $\rightarrow$ posterior **mode**. **The "estimate" you report is a statement about your loss function, not about your beliefs.**

Example 1.19 — when the loss function bites

$Y \sim$ Uniform$(0, \theta)$, default prior $p(\theta) \propto 1/\theta$. From n draws:

$$p(\theta \mid y) = \frac{n\, y_{\max}^n}{\theta^{n+1}}, \qquad \theta \geq y_{\max} \qquad \Longrightarrow \qquad E(\theta \mid y) = \frac{n}{n-1}\, y_{\max}$$

The Bayes decision under squared error loss
"Take the largest observation in your data, multiply it by $n/(n-1)$ — i.e. **slightly increase it** — and report the resulting number."

Compare: $\widehat{\theta}_{ML} = y_{\max}$, which "will underestimate θ for essentially any data set — **it will always be too low**."

Socratic prompt
The posterior did not change between the two answers. Only the loss function did.
So: is $\widehat{\theta}_{ML}$ *wrong*, or is it *right under a loss function nobody stated*? And if nobody states a loss function, what has been assumed?

Lancaster's resolution — read this one slowly

"Decision making and reporting are not necessarily alternatives. In many situations an economic model will envisage ***agents taking decisions under uncertainty****. An analysis of data using such a model will lead to you — the uncertain investigator — reporting your analysis of uncertain agents who are presumed to be taking decisions under uncertainty in an optimal (Bayesian) way.*

So really both the decision making and the reporting perspectives on Bayesian inference should receive emphasis in a text on Bayesian econometrics not because econometricians take decisions but because agents do.*"*

Key idea

This is Lancaster's answer to Giocoli's paradox. The economist *reports*; the *agent inside the model* decides. Decision theory belongs in the model, not in the estimation.

Giocoli overlay III: Wald's statistical decision problem

Wald's model has four components:

1. the available **actions**;
2. the **states of the world**, one of which is true (the parameter space);
3. the **loss function** $W(F, d)$ — the loss of taking action d when the true state is F;
4. an **experiment**, whose results depend on the true state.

A **decision function** δ associates an action with each experimental outcome.

$$r(F, \delta) = \underbrace{E[W(F, d)]}_{\text{expected loss}} + \underbrace{E[\text{cost of experimentation}]}_{\text{Wald's addition}} \qquad \text{the \textbf{risk function}}$$

Why every econometrician should recognize this

The loss "may be anything, from money to utility to lives." The statistician's task is to **choose the decision function that minimizes risk** — an optimization problem, with a cost of information, solved by an agent. *Statistics as constrained optimization.*

Giocoli (2013), "The Explorer"; Wald (1939, 1947, 1950).

Giocoli overlay III: minimax, Nature, and the complete class theorem

Minimax. Risk depends on δ (yours) and F (Nature's, unknown). Wald:

> *"Whereas the experimenter wishes to minimize the risk $r(F,\delta)$, we can hardly say that Nature wishes to maximize it. Nevertheless, since Nature's choice is unknown to the experimenter, **it is perhaps not unreasonable for the experimenter to behave as if Nature wanted to maximize the risk**."*

$\Rightarrow$ "a problem of statistical inference becomes identical with a zero sum two person game."

The complete class theorem (Wald 1947) — the punchline

The class of **Bayesian** decision functions — those based on a prior over the unknown distribution, updated by Bayes' rule — is **complete**.

For any non-Bayesian decision function there exists a Bayesian one that is at least as good at minimizing risk, for all priors.

The most rigorous frequentist decision theorist of the century proved that you never need to be non-Bayesian — and then declined to be a Bayesian.

Giocoli overlay III: Wald's instrumental Bayesianism

Wald **explicitly rejected** Bayesianism as a philosophy of probability. On introducing a prior:

> *"The reason we introduce here a hypothetical probability distribution of [states of the world] is simply that **it proves to be useful in deducing certain theorems** and in the calculation of the best system of regions of acceptance."*

And in 1950, in a footnote that deserves to be read aloud:

> *"To avoid any possibility of misunderstanding, it may be pointed out that the notions of Bayes solutions and a priori distributions are used here **merely as mathematical tools** ...and in no way is the actual existence of an a priori distribution postulated here."*

Socratic prompt

Wald used priors as scaffolding and threw them away. Lancaster (Appendix 1, point 5) reports that Bayesians return the favor — computing repeated-sampling "operating characteristics" to persuade frequentists.

Is either camp's borrowing intellectually honest? Is there any position from which both look like the same move?

Giocoli overlay III: sequential analysis, or economics invading statistics

1943. Navy Captain Garret Schuyler complains to Allen Wallis about ordnance testing:

> *"A wise and seasoned ordnance expert would see after the first few hundred rounds that the experiment need not be completed, either because the new method is obviously inferior or because it is obviously superior. ...Why do statisticians demand an experiment which is so long, so costly, and which uses up so much of your shells* ***that you've lost the war before you get the test over?****"*

Wallis and Friedman conjecture that a sequential test exists. April 1943: Wald devises the **sequential probability ratio test** — in which the number of observations is itself a random variable, and which is **explicitly founded on Bayes' rule**.

Note what just happened

The stopping rule became *endogenous*, chosen for **economic** reasons (cost of shells, cost of delay).

This is the exact situation Definition 1.2 — the likelihood principle — says you may ignore.

Giocoli overlay III: Savage takes Wald seriously

- **1949, Econometric Society.** Savage's paper: "The role of personal probability in statistics." The idea — *let's take Wald seriously*. If statistics must be behavioral, if statisticians must behave as rational economic men, we need a rigorous theory of what rational behavior *is*.
- **1951, the JASA review of Wald.** Savage introduces the **state–act–consequence** language and illustrates it with ...*whether to carry an umbrella*.
  "All problems of statistics, including those of inference, are problems of action."
- **The rhetorical move:** if a rule solves the umbrella problem, the same rule solves any statistical problem. Statistics becomes a branch of decision theory — which is to say, of economics.
- **Savage's defense of minimax:** an individual is never in complete ignorance, but a *group* might be. Minimax as group compromise — "to act so that **the greatest violence done to anyone's opinion shall be as small as possible**."

Giocoli overlay III: the fiasco, and the unintended triumph

FS's game plan: (1) build SEUT; (2) defend Wald's minimax; (3) give minimax a subjectivist reading; (4) use it to translate orthodox inference into behaviorist terms; (5) replace minimax with SEUT. Result: a wholly new statistics.

What happened: point estimation, partially. Hypothesis testing, partially. **Interval estimation: impossible.** For an agent who must *act*, accuracy estimation "could at most satisfy idle curiosity." The book ends with no conclusion at all.

- Savage (1961): "the minimax theory of statistics is an acknowledged failure."
- Savage (1972 preface): "**Freud alone could explain how [such a] rash and unfulfilled promise went unamended through so many revisions of the manuscript.**"

The unintended triumph
The failed project produced SEUT — which economics adopted wholesale. Savage was, in Lindley's phrase, **an unconscious revolutionary**. Statisticians largely ignored his Bayesianism; economists made his *agents* Bayesian instead.

Giocoli's two conjectures — why economics, and not statistics?

Conjecture 1 — consistency
Twentieth-century neoclassical economics progressively replaced **rationality as the reasoned pursuit of self-interest** with **rationality as formal consistency** (Samuelson on choice, Debreu on preferences, von Neumann on strategy). Savage extended consistency to *beliefs*. Adopting SEUT was "the inevitable, almost automatic conclusion of an intellectual journey neoclassical economists had begun long before."

Conjecture 2 — the postwar business school
The transformation of U.S. business schools created demand for a normative, teachable theory of decision making under uncertainty. SEUT fit the curriculum.

Giocoli is explicit that he has **no complete answer**. The paper's goal "was to show that the question itself is an interesting one."

Econometrics vs. decision theory — class discussion

For the room

1. Lancaster (§1.2) says econometrics is about *agents*, statistics is about *data*. Wald says statistics *is* decision theory, hence about agents too. Who is the odd one out — and what happens to §1.2's distinction if Wald is right?
2. Lancaster refuses the decision-theoretic frame because the econometrician's job is **reporting**. But a report has a format, a length, a set of things omitted. Is "just report the marginal" free of a loss function, or does it hide one?
3. Wald proved Bayes rules are a complete class and then used priors "merely as mathematical tools." Savage tried to give the tools a philosophy and failed. **Does the mathematics support any philosophy at all?**
4. Giocoli's paradox: our models assume Bayesian agents; our estimation is frequentist. Lancaster's answer: emphasize both "not because econometricians take decisions but because agents do." **Is that a resolution or a truce?**
5. If a central bank, a regulator, or a risk committee will act on your number: are you still merely reporting?

Part XI — Bridge to Ch. 2: VaR and expected shortfall

Where Chapter 2 goes

§	Topic	Why it matters to us
2.1	Methods of model checking	look again; **enlarge the model**
2.2	Informal checks: residual QQ plots	thick tails, mixtures (**returns!**)
2.3	"Uncheckable beliefs?"	what data can never touch
2.4	Formal checks: **prior predictive**	the $p(y)$ we kept dropping
2.5	**Posterior prediction**	Algorithm 2.5; the tail we care about
2.6	Posterior odds, **model averaging**	which volatility model? or all
2.7	Enlarging the model	AR(1)$\to$AR(2); homoscedastic$\to$**ARCH**

Chapter 2's opening line, which should be printed above every risk desk: **"If you don't criticize your results you can be sure that others will."**

VaR and ES are functionals of a predictive distribution

Let $\tilde{R}$ be next period's return, α the tail probability.

Definitions

$$\text{VaR}_\alpha = -q_\alpha, \quad \text{where } P(\tilde{R} \le q_\alpha) = \alpha \qquad\qquad \text{ES}_\alpha = -E[\tilde{R} \mid \tilde{R} \le q_\alpha]$$

The Bayesian question is: *which* distribution?

Plug-in: $\tilde{R} \sim p(\tilde{r} \mid \hat{\theta})$ **Predictive:** $\tilde{R} \sim p(\tilde{r} \mid y) = \int p(\tilde{r} \mid \theta) p(\theta \mid y)\, d\theta$

Recall the digression in §2.5: the plug-in "has no theoretical justification except as an approximation to (2.9) when θ is concentrated about $\hat{\theta}$."

A risk number is a quantile of the far tail. That is exactly where "approximately concentrated" stops being good enough.

Two sources of uncertainty, one number

$$\underbrace{p(\tilde{r} \mid y)}_{\text{what you should use}} = \int \underbrace{p(\tilde{r} \mid \theta)}_{\text{tomorrow's shock}} \underbrace{p(\theta \mid y)}_{\text{your ignorance about } \theta} d\theta$$

Worked case. $r_t \mid \mu, \sigma \sim n(\mu, \sigma^2)$ iid (i.e. *exchangeable* — hold that thought), vague priors. The posterior predictive is **Student-**t:

$$\tilde{R} \mid y \sim t_{n-1}\Big(\bar{r},\ s^2(1 + \tfrac{1}{n})\Big)$$

- $s^2(1 + 1/n)$: the $1/n$ is **parameter uncertainty in** μ.
- t_{n-1} rather than normal: **parameter uncertainty in** σ, and it fattens the tail.
- Plug-in reports $n(\bar{r}, s^2)$ and drops both.

Key idea
The predictive distribution is fatter-tailed than the estimated model *even when the model is exactly right*. Fat tails are not only a feature of markets; some of them are a feature of **not knowing** θ.

How much does the plug-in understate risk?

Tail multipliers, in units of s (normal model, μ and σ estimated):

n	α	VaR plug-in	VaR predictive	ES plug-in	ES predictive
25	5%	1.645	**1.745** (+6.1%)	2.063	**2.237** (+8.4%)
25	1%	2.326	**2.542** (+9.2%)	2.665	**2.978** (+11.7%)
60	1%	2.326	**2.411** (+3.6%)	2.665	**2.787** (+4.6%)
250	1%	2.326	**2.346** (+0.8%)	2.665	**2.693** (+1.1%)

- Largest **exactly where risk management is hardest**: short samples, deep tails. It grows with the horizon, the parameter count, and payoff nonlinearity (options).
- ES is corrected more than VaR: it averages where the two distributions disagree most.

t_{n-1} predictive vs. normal plug-in; verified by simulation.

Predictive VaR in Julia (Algorithm 2.5, applied)

```
using Distributions, Statistics

# posterior draws of (mu, sigma2) under the standard vague prior
nu, s2, rbar = n - 1, var(r), mean(r)
sigma2 = nu * s2 ./ rand(Chisq(nu), M)              # step 1: posterior draws
mu     = rbar .+ sqrt.(sigma2 ./ n) .* randn(M)

rtilde = mu .+ sqrt.(sigma2) .* randn(M)            # step 2: predictive draws

alpha = 0.01
q     = quantile(rtilde, alpha)
VaR   = -q
ES    = -mean(rtilde[rtilde .<= q])
```

Three lines of simulation replace a page of algebra — **and the same three lines work when the model is a GARCH, a jump diffusion, or an agent-based simulator you can only run forward** (Part VIII).

The distinction almost everyone misses

Predictive VaR — one number

$$\mathrm{VaR}_\alpha^{\mathrm{pred}}: \quad P(\tilde{R} \le -\mathrm{VaR} \mid y) = \alpha$$

Integrates out θ. This is what you *hold capital against*.

Posterior of VaR — a distribution

$$\mathrm{VaR}_\alpha(\theta) = -(\mu + \sigma z_\alpha)$$

A function of θ, so it inherits a **whole posterior** — apply it row-wise to your draw matrix (§1.4.3).

Worked example ($n = 60$ daily returns, $\alpha = 1\%$):

- plug-in VaR: 0.0247
- predictive VaR: 0.0254
- **posterior of VaR: mean** 0.0250**, 95% credible interval** $[0.0205,\ 0.0304]$

Read that interval

Your 1% VaR is known to roughly $\pm 20\%$. The regulatory report shows three decimal places and no interval at all.

Which one should a risk committee be given?

Socratic prompt

1. The predictive VaR is the honest *single* number: it already averages over your ignorance. The posterior of VaR shows how much you don't know. **Which does a capital requirement need? Which does a risk committee need?**
2. Backtesting counts exceedances against the reported VaR. Predictive VaR is calibrated for *your beliefs*; exceedance counting is a repeated-sampling operating characteristic (Appendix 1, point 5). **Is backtesting a frequentist audit of a Bayesian number — and is that incoherent, or is it exactly right?**
3. ES is a coherent risk measure and it is also **an expected loss** $E[L(\theta)]$ — Wald's risk function, wearing a Basel hat. Does the regulator's move from VaR to ES look more like Savage or more like Wald?

Model criticism for tails — why Chapter 2 comes next

Everything above assumed the model. Chapter 2 attacks it, and returns are the ideal target:

- **§2.2.1, QQ plots.** Example 2.1 is a QQ plot of 1,000 draws from a t_3 — *"the largest observations are larger than would be expected under normality and the smallest are smaller."* That is a picture of every equity return series ever plotted.
- **§2.2, Example 2.2, finite mixtures.** $p(\varepsilon) = \alpha\, n(\mu_1, \tau_1) + (1-\alpha)\, n(\mu_2, \tau_2)$ — "a good way to make Bayesian econometric modeling more flexible and less dogmatic." Calm and turbulent regimes, without saying "regime."
- **§2.7, enlarging the model.** "One might expand the model to allow for variances of the ε_t to depend upon previous realizations of the series, as in an **ARCH** model."
- **§2.6.2, model averaging.** Don't pick the volatility model — average the predictive distributions over models, weighted by posterior model probability. **Your VaR then integrates out model uncertainty too.**

The last thread: exchangeability and volatility clustering

Back to Definition 1.4. Are daily returns exchangeable in your beliefs?

- If yes, de Finetti gives you a likelihood and a prior — and iid-with-a-prior is the whole model.
- **If no** — if swapping a September 2008 return with a June 2017 return changes your joint distribution — then you have just committed to modeling the dependence. That is ARCH/GARCH/SV, and it is §2.7's "enlarging the model."

The honest statement of a Bayesian risk model

Returns are **not** exchangeable; *standardized* returns might be. So model the conditional variance, and place your exchangeability judgement one level down — on the innovations. That judgement is checkable (Ch. 2), and it is where the tail risk actually lives.

Socratic prompt

Volatility clustering is a statement about your beliefs under permutation. Is that a different claim from "volatility is persistent," or the same claim in de Finetti's grammar?

Part XII — Close

The chapter in one slide

Algorithm 1.1
1. likelihood **2.** prior **3.** insert the data **4.** Bayes' theorem **5.** criticize your model

- §1.4.1 **Likelihood** — the information transfer function. The only door the data come through. Likelihood principle: proportional likelihoods, identical inferences.
- **Exchangeability** — judge your beliefs symmetric and de Finetti hands you a likelihood *and* a prior. You were never not using one.
- §1.4.2 **Prior** — yours, revisable, encompassing; never dogmatic; conjugacy is convenience.
- §1.4.3 **Posterior** — draw it, summarize it, and above all **sample** it: "if you can sample the distribution you can know it with arbitrary accuracy."
- §1.4.4 **Decisions** — reporting and deciding, "not because econometricians take decisions but because agents do."
- $p(y)$, the denominator we dropped, is the predictive distribution — and it is where Chapter 2, ABC acceptance rates, model choice, and every risk number live.

Exercises to bring back

1. **(Appendix 1.)** Take one applied paper you admire. Rewrite its main table as a Bayesian would report it (§1.4.3). What did you have to invent? What did you have to discard?
2. **(§1.4.1.)** Build the likelihood for your own current project and write down a dataset that would embarrass it. If you cannot, say what you have assumed.
3. **(Likelihood principle.)** Simulate the binomial/negative-binomial example. Confirm the posteriors coincide. Then compute the two frequentist p-values and confirm they do not.
4. **(Exchangeability.)** For a series you care about, state the level at which you are willing to judge exchangeability. Justify it, then design a Ch. 2 check of that judgement.
5. **(ABC.)** Rerun the Part VIII sampler with $S(y) = \bar{y}$ and $\varepsilon \in \{0, 0.01, 0.05\}$. Plot the four posteriors. Explain the widening.
6. **(Prediction.)** For a return series of your choice, compute plug-in VaR, predictive VaR, and the posterior of VaR at $\alpha = 1\%$, for $n = 25,\ 60,\ 250$. Report the table. Then argue for the number you would actually put in front of a risk committee.

Sources

Primary

- Lancaster, T. (2004). *An Introduction to Modern Bayesian Econometrics.* Blackwell. Ch. 1 "The Bayesian Algorithm" (pp. 1–69); Ch. 2 "Prediction and Model Criticism" (pp. 70–111); Appendix 1 "A Conversion Manual" (pp. 359–364).
- Giocoli, N. (2013). "From Wald to Savage: *Homo Economicus* Becomes a Bayesian Statistician." *Journal of the History of the Behavioral Sciences*, 49(1).

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- Wald, A. (1939, 1945, 1947, 1950) — decision functions; SPRT; the complete class theorem.
- Savage, L. J. (1951) review of Wald, *JASA*; (1954/1972) *The Foundations of Statistics*; (1961) reassessment.
- de Finetti, B. — exchangeability and the representation theorem.
- Cox & Hinkley (1974) — conditioning and the repeated-sampling ambiguity.
- Bernardo & Smith (1994) — proof of Theorem 1.1; reference priors.

Numerical results in Parts VIII and XI were computed and verified independently of the text and are reproducible from the code shown.